

Why the diagonal–Gaussian estimator of the dark–matter scale height sits about 1σ away from the truth

Phase–space inference notes

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Abstract

When we recover the dark–matter scale height h_{dm} by back–integrating an observed phase–space snapshot and fitting a diagonal Gaussian (equivalently, a linear “ $au + b$ ” bijection) to the initial cloud, the profile–likelihood minimum lands a few parsec away from the true value—for our fiducial $N = 20\,000$, seed–7 data set it sits at $h \simeq 258\text{ pc}$ versus a truth of 250 pc . This note shows that the offset is *not* a systematic bias, an integrator artifact, or a modelling error: it is the ordinary $\sim 1\sigma$ statistical scatter of the estimator. We derive the error bar $\sigma_h = \sigma_\rho/|d\rho/dh| \approx (1/\sqrt{N})/|d\rho/dh|$ by propagating the sampling noise of a correlation coefficient through the estimating equation, and verify it reproduces the measured seed–to–seed scatter of 8 pc .

1 The observation

The data are N stars drawn from a centred, separable initial distribution function $f_0(z_0, w_0) = \mathcal{N}(0, \sigma_z^0) \mathcal{N}(0, \sigma_w^0)$, evolved for $t_{\text{obs}} = 400\text{ Myr}$ in the true potential (scale height $h_{\text{true}} = 250\text{ pc}$), with no satellite. To infer h we back–integrate the snapshot to $t = 0$ under a trial potential of scale height h , obtaining a cloud $(z_0, w_0) = T_h(z, w)$, and fit a *diagonal* Gaussian to it. The negative log–likelihood of that model depends only on the marginal widths,

$$\text{NLL}(h) = N \left[\ln(2\pi) + \ln \sigma_z(h) + \ln \sigma_w(h) + 1 \right], \quad (1)$$

so the estimator $\hat{h} = \arg \min_h \text{NLL}(h)$ minimises the product $\sigma_z(h) \sigma_w(h)$. Empirically the minimum sits a few pc above h_{true} , and it does so *reproducibly* across our tests—simply because those tests reuse the same random seed.

2 What the estimator really measures

Write the 2×2 covariance of the back–integrated cloud as

$$\text{Cov}(h) = \begin{pmatrix} \sigma_z^2 & \rho \sigma_z \sigma_w \\ \rho \sigma_z \sigma_w & \sigma_w^2 \end{pmatrix}, \quad \det \text{Cov}(h) = \sigma_z^2 \sigma_w^2 (1 - \rho^2), \quad (2)$$

where $\rho(h)$ is the z – w correlation coefficient. Hence

$$\ln(\sigma_z \sigma_w) = \frac{1}{2} \ln \det \text{Cov}(h) - \frac{1}{2} \ln(1 - \rho(h)^2). \quad (3)$$

The back-integration T_h is a *symplectic* map, so it conserves the phase-space measure. For these near-harmonic clouds the second-moment area $\det \text{Cov}(h)$ is conserved to excellent accuracy as well (numerically it is flat to $< 0.1\%$ across the scanned range, $\det \text{Cov} \simeq 2.645 \times 10^7$). With the first term in (3) constant, minimising $\ln(\sigma_z \sigma_w)$ is *equivalent to* maximising $1 - \rho^2$, i.e. to the condition

$$\boxed{\hat{\rho}(\hat{h}) = 0} \quad (4)$$

The diagonal model, which ignores the correlation, therefore places its optimum exactly at the scale height for which the recovered cloud is *uncorrelated*.

3 Why $\rho = 0$ is not exactly at the truth

At the true scale height the map $T_{h_{\text{true}}}$ inverts the forward evolution and returns the original cloud, whose *population* correlation is zero. But the *sample* correlation of a finite draw is not exactly zero: for N points with true correlation ≈ 0 it fluctuates about 0 with a spread σ_ρ . Thus

$$\hat{\rho}(h_{\text{true}}) = \underbrace{\rho_{\text{true}}(h_{\text{true}})}_{=0} + \delta\rho, \quad \mathbb{E}[\delta\rho] = 0, \quad \text{SD}[\delta\rho] = \sigma_\rho. \quad (5)$$

Because $\hat{\rho}(h_{\text{true}}) \neq 0$ in any single realisation, the zero of $\hat{\rho}(h)$ —and hence $\hat{h}$ —is displaced from h_{true} .

4 The error bar: propagating σ_ρ through the estimating equation

This is the delta method (equivalently, the implicit-function theorem) applied to (4). Linearise $\hat{\rho}(h)$ about the truth,

$$\hat{\rho}(h) \approx \hat{\rho}(h_{\text{true}}) + \frac{d\rho}{dh}(h - h_{\text{true}}), \quad (6)$$

and impose $\hat{\rho}(\hat{h}) = 0$:

$$\frac{d\rho}{dh}(\hat{h} - h_{\text{true}}) + \delta\rho \approx 0 \quad \implies \quad \hat{h} - h_{\text{true}} \approx -\frac{\delta\rho}{d\rho/dh}. \quad (7)$$

Taking the standard deviation of (7) gives the central result,

$$\boxed{\sigma_h = \frac{\sigma_\rho}{|d\rho/dh|}} \quad (8)$$

This is the generic standard error of a root of an estimating equation $g(\hat{\theta}) = 0$, namely $\text{SE}(\hat{\theta}) = \text{SD}[g]_{\text{truth}}/|g'|$; it is the same “output error = input error / sensitivity” one uses for any indirect measurement.

The $1/\sqrt{N}$ factor. The remaining ingredient is the sampling scatter of a correlation coefficient. Fisher’s classical result for the sample correlation $\hat{\rho}$ of a bivariate normal sample is

$$\text{Var}(\hat{\rho}) \approx \frac{(1 - \rho^2)^2}{N} \xrightarrow{\rho \rightarrow 0} \frac{1}{N}, \quad \text{so} \quad \sigma_\rho \approx \frac{1}{\sqrt{N}}. \quad (9)$$

Substituting into (8),

$$\sigma_h \approx \frac{1/\sqrt{N}}{|d\rho/dh|}. \quad (10)$$

N enters only here: more stars pin ρ tighter, but the small slope $d\rho/dh$ —the weak, anharmonic grip of h on the data—sets a large lever arm that inflates the resulting uncertainty in h .

5 Numerical verification

For the fiducial data ($N = 20\,000$, forward integration with 12 000 steps) the decomposition (3) is confirmed directly: $\det \text{Cov}$ is flat, $\rho(h)$ marches monotonically through zero, and the $\ln(\sigma_z \sigma_w)$ minimum coincides with the $\rho = 0$ crossing (Table 1).

h [pc]	σ_z [pc]	ρ	$\det \text{Cov}$
230	258.1	+0.033	2.646×10^7
250	257.4	+0.010	2.645×10^7
259	257.2	≈ 0	2.645×10^7
270	257.2	−0.011	2.645×10^7

Table 1: The correlation $\rho(h)$ passes through 0 near $h \simeq 259$ pc, while $\det \text{Cov}$ is essentially constant. The diagonal–Gaussian minimum tracks the $\rho = 0$ point, not the truth.

Repeating the fit over independent random seeds shows the offset is zero–mean and of the size predicted by (8) (Table 2). The measured slope is $|d\rho/dh| \approx 0.0011 \text{ pc}^{-1}$, and the sample scatter of the initial–cloud correlation across seeds is $\sigma_\rho = 0.0088$ (versus the ideal $1/\sqrt{20\,000} = 0.0071$; the excess comes from the evolve/back–integrate round trip mildly amplifying the noise). Equation (8) then predicts

$$\sigma_h = \frac{0.0088}{0.0011} \approx 8.0 \text{ pc}, \quad (11)$$

in close agreement with the directly measured seed–to–seed scatter of 8.4 pc. The fiducial seed–7 result, $\hat{h} = 259$ pc, is thus $+1.1\sigma$ —an entirely ordinary draw.

	ρ_{IC}	$\hat{h} - h_{\text{true}}$ [pc]	
seed 7 (fiducial)	+0.0100	+9.4	
seed 1	+0.0036	+3.7	
seed 3	−0.0012	−0.9	
seed 2	−0.0130	−11.9	
seed 6	−0.0130	−12.6	
mean (8 seeds)	≈ 0	−3.5	(consistent with 0)
scatter (8 seeds)	0.0088	8.4	$\approx \sigma_h$
$N = 80\,000$ scatter	0.0038	2.4	$\propto 1/\sqrt{N}$

Table 2: The minimum offset tracks the finite–sample initial–cloud correlation ρ_{IC} , averages to zero, and shrinks as $1/\sqrt{N}$ (scatter 8.4 pc $\rightarrow$ 2.4 pc from $N = 20\,000$ to 80 000).

6 Conclusion

The few–parsec displacement of the profile minimum is the 1σ statistical error of the estimator, not a bias:

1. the diagonal/affine model reads h only through the condition $\hat{\rho}(h) = 0$;
2. in a finite sample $\hat{\rho}(h_{\text{true}}) \neq 0$ by $\mathcal{O}(1/\sqrt{N})$;
3. the shallow slope $d\rho/dh$ converts that tiny correlation error into a displacement $\sigma_h = \sigma_\rho/|d\rho/dh| \approx (1/\sqrt{N})/|d\rho/dh|$.

For our data this is ≈ 8 pc at $N = 20\,000$, so the fiducial $\hat{h} = 258$ pc should be read as $h = 250 \pm 8$ pc. The uncertainty is large not because the data are few but because each star carries little information about h (the anharmonic insensitivity encoded in the small $d\rho/dh$); a richer estimator that uses more than the correlation could tighten it.